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ER Model Requirements

EGEN5208W Databases for Soft Engineers

Course: EGEN5208W Databases for Soft Engineers

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Individual Project Requirements

Your ER model must satisfy the following requirements. The ER model is evaluated as a conceptual design artifact, not as an implementation.

The ER model must include at least six entities and at least five relationships. All entities and relationships included in the diagram must be necessary to support the required application functionality. Adding additional entities or relationships beyond what is required does not increase marks.

The ER model must be consistent with the functional requirements, the database schema, and the implemented SQL. Inconsistencies between these artifacts will be treated as design errors.

Required Deliverable

You must submit one ER diagram drawn using either Chen notation or a UML-like notation. The chosen notation must be used consistently throughout the diagram.

The ER diagram must be clear, professional, and readable. It must show the complete conceptual structure of the system as described by the project requirements.

1. Core Principle

The ER model defines the conceptual structure of the system's data.

The ER model must be sufficient to explain the data requirements of the system. Any functionality implemented in the application or database must be representable using the ER model. If required data or relationships are not captured in the ER model, the design will be considered incomplete.

2. ER Model Content Requirements

2.1 Entities

The ER model must include at least six entities. Each entity must represent a meaningful real-world concept relevant to the system.

Entities must have clearly identified primary keys. Attributes included in an entity must describe that entity only. Artificial or placeholder entities created solely to increase the entity count are not acceptable.

2.2 Attributes

All relevant attributes must be shown in the ER diagram. Primary keys must be clearly indicated. Attribute naming should be clear and meaningful. Derived or computed attributes should not be included unless their presence is explicitly justified.

2.3 Relationships

The ER model must include at least five relationships. Relationships must be clearly named and must reflect meaningful associations required by the system. Each relationship must indicate cardinality. Relationships that are not required to support the system's functionality should not be included.

3. Alignment with Functional Requirements

The ER model must support all eight required application operations.

It must be possible to explain, using the ER model, how the data required for each operation is represented. If an operation requires data that is not captured in the ER model, the ER model is considered incomplete.

4. ER Diagram Format and Notation

The ER diagram must be presented in a clear and professional format.

It must include: all entities, all relevant attributes, all relationships, and cardinality information for relationships.

The diagram must use either Chen notation or UML-like notation. Mixing notations is not permitted.

Any standard diagramming tool may be used. Hand-drawn or unclear diagrams will not be accepted.

5. General Design Expectations

The ER model must be internally consistent and conceptually sound. It must be completed before database schema design or application implementation. The ER model must describe the system at a conceptual level and should not include implementation-specific details.

[Next: Database Schema & SQL Requirements →](#)